

ISINGOMA ARTHUR

Lubowa, Seguku, Uganda

BSc COMPUTER SCIENCE STUDENT

(IoT | Enterprise Networking | Database Systems | Software Development)

Experience Summary

Objective

Motivated Computer Science student at Uganda Christian University (UCU), passionate about building intelligent systems that solve real-world problems. Skilled in network design, IoT prototyping, database systems, and software development. Demonstrated ability to design enterprise-level multi-site network architectures, develop IoT-driven agricultural monitoring systems, and deliver academic projects across data structures, discrete mathematics, and digital electronics. Committed to applying technology to the Ugandan agricultural and infrastructure sectors.

Skills Summary

- Programming Languages: Python (80%), SQL (75%), HTML/CSS (70%), JavaScript (55%)
- Networking & Systems: OSPF / VLAN Design, Cisco Packet Tracer, VirtualBox & VM Management, HSRP, MPLS/IPSec WAN, 802.1X Authentication
- Database Systems: MySQL, database normalisation (1NF-3NF), schema design, views, constraints, ERD modelling
- IoT & Embedded Systems: Sensor integration, agricultural IoT prototyping (FarmGuard / FarmSense), NARO Mukono field research
- Tools & Platforms: SolidWorks / CAD (3D modelling & parametric design), LaTeX (technical report writing), Blender, VS Code
- Software Development: Data structures (arrays, linked lists, stacks, queues, BSTs), discrete mathematics, digital electronics
- Design Thinking: MITRE Prototyping, wireframing, ERD design, Feedback Capture Grids, challenge statement frameworks
- Communication: Technical report writing, LinkedIn content, Medium blog articles, academic presentations

Project Experience (1)

IoT Developer & System Designer			
Project Name	FarmGuard / FarmSense — Smart Agricultural Monitoring	Team Size	Group 21
Institution	Uganda Christian University, Mukono	Period	2025 – Present
Project Description	As part of Group 21 at UCU, designed and prototyped a smart IoT-driven agricultural monitoring system targeting Ugandan farming challenges. The system integrates soil and weather sensors with a mobile application to deliver real-time alerts, agronomist recommendations, and farm reports. Conducted field research at NARO Mukono with farmer interviews to validate system design. Project encompasses full-stack design thinking from challenge statement to functional prototype.		

Role & Contribution	- Designed an 8-entity ERD (Farmer, Admin, Technician, Agronomist, Sensor, Sensor_Reading, Alert, Farm_Report) for the FarmSense database backend - Built interactive app wireframes inside a phone mockup, mapping user flows for farmers and agronomists - Completed MITRE Prototyping worksheets and hardware/app flowcharts to document system architecture - Conducted field visits to NARO Mukono with structured farmer interviews to gather real-world requirements - Produced a Feedback Capture Grid from a design review session and iterated on system design accordingly - Documented project findings in a LaTeX report and published a LinkedIn post covering the field visit - Applied design thinking methodologies: challenge statement diagrams, priority matrix, problem statements
Technology & Tools	- IoT & Sensors: Soil moisture sensors, weather data modules, real-time alert systems - Database: MySQL, ERD design, normalisation, relational schema modelling - Documentation: LaTeX, wireframing tools, MITRE prototyping framework - Research: Field interviews, observational data collection, NARO Mukono partnership
Key Achievements	Still in the making

Project Experience (2)

Network Engineer — Enterprise Network Design			
Project Name	UCU Digital Services Ltd — Multi-Site Enterprise Network	Team Size	Rack 4
Institution	Uganda Christian University, Mukono	Period	2025 – Present
Project Description	Designed a fully redundant multi-site enterprise network for UCU Digital Services Ltd, connecting headquarters in Mukono to branch offices in Kampala, Mbarara, and Jinja. The design incorporates OSPF hierarchical area routing, a large VLAN range for network segmentation, MPLS/IPSec WAN links, dual firewalls, HSRP for gateway redundancy, and 802.1X port authentication. Emphasis on fault tolerance, security, and scalability across all sites.		

Role & Contribution	▪ Designed OSPF hierarchical area topology with route summarisation to reduce LSA flooding across WAN links ▪ Planned and implemented VLAN segmentation strategy covering all departments across 4 sites ▪ Configured HSRP for default gateway redundancy on core distribution switches ▪ Implemented dual firewall architecture and 802.1X authentication for enhanced perimeter security ▪ Produced enhanced network topology diagrams with fault tolerance and redundancy improvements ▪ Analysed OSPF scalability trade-offs and documented hierarchical design decisions in technical reports ▪ Simulated network configurations using Cisco Packet Tracer for validation
Technology & Tools	▪ Routing & Switching: OSPF (multi-area), VLAN, HSRP, STP, inter-VLAN routing ▪ WAN Technologies: MPLS, IPSec VPN tunnelling ▪ Security: Dual firewalls, 802.1X port-based authentication, ACLs ▪ Simulation: Cisco Packet Tracer, network topology diagramming tools
Key Achievements	Successfully delivered a fully redundant, secure enterprise network design for a 4-site organisation with comprehensive fault tolerance, OSPF scalability planning, and documented topology diagrams.

Professional Certifications / Trainings

▪ SolidWorks 3D Modelling Workshop — Trinity 2026 (4-day course: parametric modelling, FeatureManager, digital twins, pulley assembly, ISO drawings) ▪ Cisco Packet Tracer — Network simulation and enterprise topology design (in progress) ▪ LaTeX Technical Writing — Academic and professional documentation ▪ Additional certifications planned: Cisco CCNA, CompTIA Network+ (2026+)
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Educational Qualification

Education & Credentials	Bachelor of Science in Computer Science (BSc CS) Uganda Christian University (UCU) — Mukono, Uganda Faculty of Engineering, Design & Technology Group 21 2025 – Present Uganda Advanced Certificate of Education (A-Level) St. Mary's College Kisubi (SMACK) — Kisubi, Uganda 2023 – 2024
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Referees

UCU Instructor	Mr. Charles Galiwango — Instructor, Faculty of Engineering, Design & Technology, UCU
Contact	Email: isingomaarthur30@gmail.com Phone: 0741 897 680 Location: Lubowa, Seguku, Uganda University: UCU, Mukono

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